

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 9

Tutorial exercises

1. If the incident and deflected wave vectors of a particle are $\vec{k}$ and $\vec{k}'$ respectively by a potential $V(\vec{r})$ then the Born approx. for the scatted amplitude is given by

$$f_B(\vec{k}, \vec{k}') = -\frac{\mu}{2\pi\hbar^2} \int d\vec{r} e^{-i\vec{k}' \cdot \vec{r}} V(\vec{r}) e^{i\vec{k} \cdot \vec{r}} \quad (1)$$

where μ is the reduced mass. Now for a central potential it becomes

$$f_B(q) = -\frac{2\mu}{\hbar^2 q} \int_0^\infty dr \sin(qr) r V(r) \quad (2)$$

where we consider elastic scattering for which $q = |\vec{k} - \vec{k}'| = 2k \sin(\theta/2)$ where θ is angle of deflection.

Using this calculate the differential scattering cross section from the Yukawa potential

$$V_Y(r) = V_0 \frac{e^{-\alpha r}}{\alpha r} \quad (3)$$

Then calculate the total scattering cross section.

Home exercise

2. Show that the s -wave phase shift for a square well of depth V_0 and range r_0 is

$$\delta_0 = -kr_0 + \tan^{-1} \left(\frac{k}{k'} \tan(k'r_0) \right) \quad (4)$$

where k' and k are the wave numbers inside and outside the well.

3. A particle of mass m is scattered from a spherical repelling potential of region r_0 , given as

$$V(r) = V_0 \quad r \leq r_0 \quad (5)$$

$$= 0 \quad r > r_0 \quad (6)$$

- (a) Using Born approx. calculate the total cross-section σ_T in the limit of low energies.
- (b) Calculate σ_T using partial wave expansion considering only the contribution from s -wave.